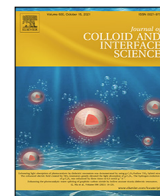




Contents lists available at ScienceDirect

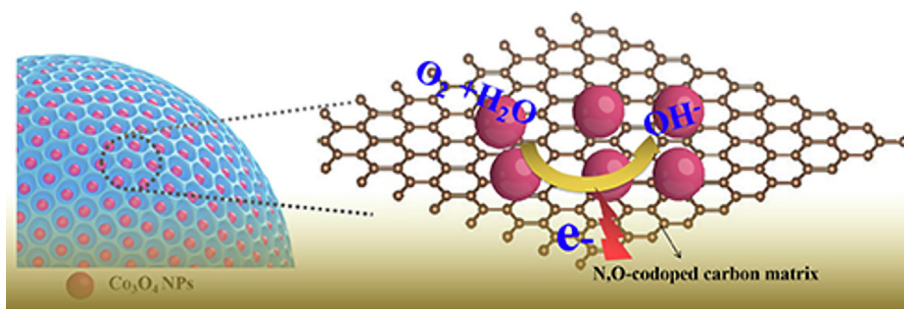
Journal of Colloid and Interface Science

journal homepage: www.elsevier.com/locate/jcis

N,O-codoped carbon spheres with uniform mesoporous entangled Co_3O_4 nanoparticles as a highly efficient electrocatalyst for oxygen reduction in a Zn-air battery

Yan Liu^a, Tao Zhang^b, Yu E. Duan^c, Xin Dai^a, Qiang Tan^a, Yuanzhen Chen^a, Yongning Liu^{a,*}^a State Key Laboratory for Mechanical Behavior of Materials, Xi'an Jiaotong University, Xi'an 710049, PR China^b Department of Chemical and Biochemical Engineering, Rutgers, The State University of New Jersey, Piscataway, NJ 08854, USA^c School of Chemistry and Chemical Engineering, Xi'an University of Science and Technology, Xi'an 710054, PR China

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 5 April 2021

Revised 6 July 2021

Accepted 7 July 2021

Available online 10 July 2021

Keywords:

Entangled Co_3O_4

N,O-codoped carbon spheres

Oxygen reduction reaction

Zn-air battery

ABSTRACT

Highly efficient electrochemical catalysts for oxygen reduction reactions (ORRs) are urgently needed for various energy conversion and storage devices to overcome sluggish ORR kinetics. Here, N,O-codoped carbon spheres with uniform mesopores and a high specific surface area were used as supports for decorating Co_3O_4 nanoparticles via a facile immersion route. In addition to the benefit of ions and gas mass transfer, the abundant mesopores present in the three-dimensional (3D) carbon spheres also confine and isolate the Co_3O_4 nanoparticles growing in it, which help to provide rich Co_3O_4 active sites. The resulting hybrid material exhibits superior ORR activity in terms of even-better half-wave potential and stability than that of commercial Pt/C (40 wt%) in 0.1 M KOH electrolyte. To verify its catalytic activity, the hybrid material was employed as the cathode catalyst in a flexible solid-state zinc-air battery, which achieves a high power density of 227 mW cm^{-2} ; this power density is much higher than that of a Pt/C catalytic zinc-air battery (133 mW cm^{-2}) under identical conditions. The improvement in catalytic activity in both aqueous and nonaqueous electrolytes can be attributed to the abundant active sites of the entangled Co_3O_4 nanoparticles, as well as the novel N,O-codoped carbon structure.

© 2021 Elsevier Inc. All rights reserved.

1. Introduction

To overcome the problems of environmental pollution and the energy crisis, energy storage and conversion technologies, such as fuel cells and metal-air batteries, have been extensively investi-

* Corresponding author.

E-mail address: ynliu@xjtu.edu.cn (Y. Liu).

gated for their low cost, environmental friendliness and good energy density [1–4]. However, as a main component of the above devices, the sluggish oxygen reduction reaction (ORR), an uphill reaction that contains a multistep electron transfer, restricts the practical application of these devices. Noble metal platinum (Pt)-based catalysts are currently considered the most promising materials for the ORR. However, their high price, scarcity and inferior durability have prohibited their large-scale application [5,6]. Therefore, much effort has been devoted to exploring nonprecious and highly efficient catalysts.

Inexpensive heteroatom-doped carbon materials with excellent electrocatalytic activity have been discovered and demonstrated to be promising alternative materials to replace Pt/C. It has been reported that doped heteroatoms including nitrogen (N) [7–9], sulfur (S) [10–12], boron (B) [13,14], phosphorus (P) [15,16], etc. can modify the electron structure and tailor charge density of carbon nanostructures, thus boosting the chemical adsorption/desorption of intermediates and accelerating the ORR rate [17–19].

Nonprecious metal materials, such as metal oxides [20–22], metal nitrides [23,24] and metal sulfides [25,26], are also prospective candidates to replace precious Pt/C materials due to their low price, good activity and stability. Among the inexpensive metal catalysts, cobalt oxide (Co_3O_4) is one of the most promising electrocatalysts for the ORR because of its abundance, high ORR activity and superior stability [27–29]. The mixed valence of Co^{2+} and Co^{3+} in Co_3O_4 can lead to electron donor–acceptor active site pairs during oxygen absorbance in electrochemical reactions [30]. However, pure Co_3O_4 generally has unacceptable electrical conductivity and low exposure of active sites, thus giving low catalytic activity [31,32]. In addition, metal-based nanoparticles invariably suffer from aggregation during preparation and long-term operation due to their small particle size and high surface energy, thus leading to a decrease in available ORR active sites [33]. Although the performance of such catalysts has been significantly improved, it is still far from satisfactory.

To further improve the ORR activity, tremendous efforts have been made to combine heteroatom-doped carbons and metal materials. The generated synergistic effects between the carbon support and metals can remarkably promote the ORR performance [34,35]. For instance, Co/CoN_x nanoparticle-embedded N-doped carbon exhibits an improved half-wave potential ($E_{1/2}$) of 0.87 V vs. reversible hydrogen electrode (RHE) in alkaline media, which is much better than each single part but still slightly negative than the value of 0.97 V vs. RHE for Pt/C [36]. In another example [37], Co_3O_4 nanosheet clusters anchored on N-doped carbon nanotube/3D graphene exhibit an $E_{1/2}$ of 0.8 V vs. RHE. This result is more positive than each single part but still more negative than that of commercial Pt/C in alkaline electrolyte. Clearly, the combination of metal-based materials together with carbon catalysts can predominantly boost the ORR performance but also shows that there is still room to improve the activity of the hybrid.

Generally, carbon nanotubes and graphene doped with heteroatoms are the most widely used materials to anchor metal materials [38,39] because of their excellent electronic conductivity, good mechanical integrity and low cost. Herein, a nitrogen (N) and oxygen (O)-codoped carbon sphere (NOCS) with uniformly abundant mesopores, which has been reported in our previous work [40], is introduced as a new support to entangle Co_3O_4 nanoparticles (Co_3O_4 NPs). This novel three-dimensional (3D) nanomaterial with a high specific surface area and uniform pore diameter (~12 nm) facilitated ion/air diffusion transfer paths and exhibited good ORR performance. After growing Co_3O_4 NPs inside the pores of the N,O-codoped carbon spheres, the resulting material displays an ORR performance that is further enhanced than any individual part. The activity is found to be even better than the benchmark of commercial Pt/C in alkaline electrolyte solution. A primary

solid-state zinc-air battery was assembled with the prepared material as an air electrode to further evaluate its ORR activity, which shows a satisfactory power density and remains stable for a long time during electrocatalysis.

2. Experimental section

2.1. Materials and reagents

Colloidal silicas (Ludox HS-30, 30%), ammonium persulfate ($(\text{NH}_4)_2\text{S}_2\text{O}_8$, AR, $\geq 98\%$), sodium hydroxide (NaOH, AR, $\geq 96.0\%$) and potassium hydroxide (KOH, AR, $\geq 90.0\%$) were purchased from Sigma-Aldrich. Aniline ($\text{C}_6\text{H}_7\text{N}$, AR, $\geq 99.5\%$) and cobalt nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, AR, $\geq 98.5\%$) were purchased from Sino-pharm Chemical Reagent Co., Ltd (Shanghai, China). N, N-methylene bisacrylamide ($\text{C}_7\text{H}_{10}\text{N}_2\text{O}_2$, AR, $\geq 99.0\%$) and acrylic acid ($\text{C}_3\text{H}_4\text{O}_2$, AR, $\geq 99.5\%$) were received from Kemiou Chemical Reagent Co., Ltd (Tianjin, China). Pt/C (HiSPEC 4000, 40%) was obtained from Johnson Matthey. Polytetrafluoroethylene solution (PTFE, 60%) was provided by 3F New Materials Co., Ltd (Shanghai, China). Gas diffusion layers were purchased from Sipulin New Energy Technology Co., Ltd (Changsha, China). All reagents were used as received.

2.2. Synthesis of Co_3O_4 -entangled N,O-codoped carbon spheres

N,O-codoped carbon spheres were prepared using a colloidal silica (SiO_2) suspension as hard templates following a modified method described in our previous report [40]. Carbon precursors were first synthesized, which was then subjected to pyrolysis under a nitrogen (N_2) atmosphere in a temperature-programmable tube furnace as follows. The temperature of the furnace was raised to 300 °C at a heating rate of 1 °C min^{-1} and kept for 3 h. It was then increased to 900 °C at a heating rate of 10 °C min^{-1} and aged for 2 h. The embedded SiO_2 templates in the carbonized product were etched away by NaOH [40]. The resulting material was denoted as N,O-codoped carbon spheres (NOCS).

To entangle Co_3O_4 into NOCS, the NOCS powder was ultrasonically dispersed into 40 mL distilled water. Then, 0.5 M $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ aqueous solution was added dropwise into the above solution under magnetic stirring. The received mixture was stirred for 6 h and then vacuum filtered to obtain the solid powder. The material was dried at 80 °C overnight in an oven and then subjected to aging at 300 °C for 2 h at a heating rate of 3 °C min^{-1} under a N_2 atmosphere. The resulting material is named $\text{Co}_3\text{O}_4/\text{NOCS}$. Nanotube-supported Co_3O_4 ($\text{Co}_3\text{O}_4/\text{CNTs}$) was prepared using the same procedures except that the NOCS was replaced by CNTs. The $\text{Co}_3\text{O}_4/\text{NOCS}$ was immersed in a 0.5 M H_2SO_4 acid solution for 12 h under magnetic stirring at room temperature to obtain acid-treated $\text{Co}_3\text{O}_4/\text{NOCS}$.

2.3. Synthesis of solid-state electrolyte

The solid-state electrolyte was prepared as the followings. First, solution A and B was prepared by dissolving 18 g KOH in 26 mL distilled water and dispersing 0.5 g N, N-methylene bisacrylamide in 3 g acrylic acid, respectively. Subsequently, solution A and B was mixed together under magnetic stirring for 10 min before filtering to obtain a relative clear liquid. Finally, three drops of 5 wt% ammonium persulphate solution were added into the above liquid and the polymer gel electrolyte was obtained.

2.4. Characterization of catalysts

X-ray diffraction patterns (XRD, X'Pert PRO) was conducted to elucidate the crystal structure of the catalysts with Cu K α radiation as X-ray source. Morphologies of the materials and the solid-state gel were revealed by a field-emission scanning electron microscope (FE-SEM, JSM-6700F, JEOL, Japan). Transmission electron microscope (TEM) and high-resolution TEM (HRTEM) (JEM-200CX, NEC, Japan) were employed to investigate the microstructures of the materials. The specific BET (Brunauer-Emmett-Teller) surface areas, BJH (Barrett-Joyner-Halenda) pore sizes, and pore volumes of the materials were measured by ASAP 2460 gas porosimeter (Micromeritics). The thermogravimetric (TG) curves of the samples were obtained using a Setaram Labsys Evo (France). The composition of the materials, including elements and chemical state, were determined by an X-ray photoelectron spectroscopy (XPS) using an ESCALAB Xi⁺ instrument (Thermo Fisher Scientific) operating with Al K α as the X-ray source. The binding energies were calibrated with respect to the C 1s peak at 284.6 eV. The oxygen and nitrogen contents were determined by rapid OXY cube and vario EL cube elemental analyzers (the products of Elementar company, Germany), respectively.

2.5. Electrochemical measurements and preparation of electrodes

Electrochemical measurements were conducted on electrochemical workstation (CHI 750D, CH Instruments) with a three-electrode cell, which contains a Hg/HgO (1 M KOH) reference electrode, a carbon counter electrode, and a glassy carbon-based rotating disk electrode (RDE) or rotating ring-disk electrode (RRDE) coated with catalyst as a working electrode (RDE: 5 mm diameter, 0.196 cm²; RRDE: a Pt ring (6.5 mm inner diameter and 7.5 mm out diameter, 0.192 cm²) together with a glassy carbon-based RDE (5 mm diameter, 0.196 cm²); Pine Research Instrumentation). The reference electrode was calibrated with respect to the reversible hydrogen electrode (RHE) by immersing a calibration tube containing a Pt wire in a H₂ gas-purged KOH solution (0.1 M) [41].

The working electrode was prepared by coating the catalyst suspension onto the RDE or the RRDE as follows. First, 2 mg of each catalyst was dispersed in a solution composed of ethanol (0.25 mL), distilled water (0.1 mL) and Nafion solution (5 wt%, 10 μ L) by sonicating for 1 h. The homogeneous black ink (10 μ L) was taken and deposited onto the surface of a glassy carbon electrode and dried naturally. The loading of the catalyst was calculated to be 0.283 mg cm⁻². Pt/C (40 wt%) was employed for a comparison using the following ink ratios: 2 mg Pt/C was well distributed by ultrasonical dispersion in a mixture solution containing of 2 mL ethanol and 20 μ L Nafion solution (5 wt%, 10 μ L). The loading of Pt/C was 0.0756 mg cm⁻².

Cyclic voltammetry (CV) curves were obtained in O₂- and N₂-saturated KOH solutions (0.1 M) with a sweep rate of 50 mV s⁻¹. Linear scan voltammetry (LSV) was performed at a scan rate of 10 mV s⁻¹ in O₂-saturated KOH solution (0.1 M) using RDE at different speed rates or using RRDE with a spinning rate of 1200 rpm. The stability of the catalysts (chronoamperometric responses) and the changes in current density before and after the addition of methanol were measured in O₂-saturated KOH solution (0.1 M) at a rotating rate of 1200 rpm. The experiments were carried out at room temperature.

The electron transfer number (n) involved in ORR is calculated based on Koutechy-Levich (K-L) equation [42]:

$$\frac{1}{j} = \frac{1}{j_k} + \frac{1}{B\omega^{1/2}} \quad (1)$$

$$B = 0.62nFD_0^{2/3}v^{-1/6}C_0 \quad (2)$$

Where j is the measured current density, j_k is the kinetic current density, ω is the angular velocity (rad), n is the electron transfer number, F is the Faraday constant (96500C mol⁻¹), D_0 is the diffusion coefficient of O₂ in 0.1 M KOH (1.9×10^{-5} cm² s⁻¹), C_0 is the bulk concentration of O₂ (1.2×10^{-6} mol cm³), and v is the viscosity of the electrolyte (1.0×10^{-2} cm² s⁻¹).

During the RRDE test, the ring potential was fixed with a constant value of 1.5 V vs. RHE. The calculations of % of H₂O₂ and electron transfer number (n) are made based on the values of I_R and I_D measured with RRDE and using Eq. (3) and (4) [43]:

$$\%H_2O_2 = \frac{200 \times \frac{I_R}{N}}{I_D + \frac{I_R}{N}} \quad (3)$$

$$n = 4 \times \frac{I_D}{\frac{I_R}{N} + I_D} \quad (4)$$

Where N is the efficiency of current collection at the ring electrode (0.38). I_R is the ring current. I_D is the disk current.

The j_k in the Tafel plots of E vs. $\log(j_k)$, which represents the kinetic current density, was calculated from the mass-transport correction of RDE [44]:

$$j_k = \frac{j \times j_L}{j_L - j} \quad (5)$$

Where j is the measured current density, j_L is the diffusion-limiting current density.

The swelling capacity measurement of the polymer gel was tested by soaking the gel into a 4 M KOH solution and studying the change of gel weight as the time extended. The ionic conductivity of the gel was measured using Eq. (6) [44]:

$$\delta = \frac{L}{(R \times A)} \quad (6)$$

Where δ denotes the ionic conductivity, L is the thickness of gel, R is the bulk resistance of the gel, and A is the area in the transverse direction. The bulk resistance of the gel was measured by the impedance spectroscopy over a frequency range of 0.01 ~ 10⁵ Hz on an electrochemical workstation (CHI 750D, CH Instruments). A piece of gel with a size of 2.5 cm long, 1 cm width and 0.5 cm thick was used to test the resistance.

2.6. Preparation of Zn-air battery

To prepare the air electrode, a gas diffusion layer was first pressed together with the Cu foam at a pressure of 4 MPa. 75 wt% catalyst and 25 wt% polytetrafluoroethylene (PTFE, 30 wt% solution) with enough ethanol was then uniformly dispersed via ultrasonication for 30 min. The obtained slurry was coated onto the other side of Cu foam (without gas diffusion layer) with a catalyst loading of 12 mg cm⁻². The Cu foam coated with catalyst was pressed again under a pressure of 4 MPa after drying in an oven at 60 °C for 4 h.

Finally, the Zn foil, polymer gel and air electrode were assembled into a battery. It should be noted that the catalyst side of the air electrode should face to the polymer gel. The performance of the battery was evaluated by performing discharge and stability tests using a battery testing system (Neware Technology Co., Ltd., Shenzhen, China).

3. Results

3.1. Preparation and characterization of Co₃O₄/NOCS

The preparation of Co₃O₄ NPs supported on N,O-codoped carbon spheres is illustrated in Fig. 1a. First, N,O-codoped carbon

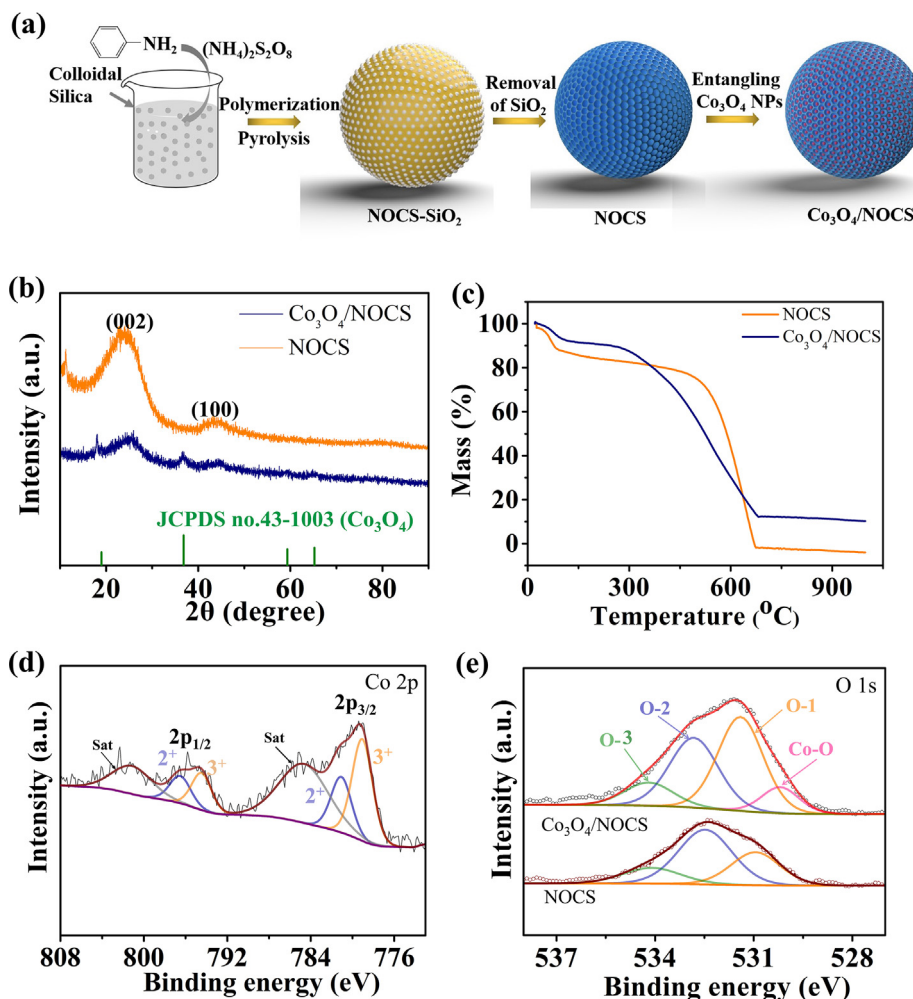


Fig. 1. (a) Preparation process for the $\text{Co}_3\text{O}_4/\text{NOCS}$ material. (b) XRD patterns of NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$. (c) TG curves of NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$. (d) Deconvoluted Co 2p XPS peaks of the $\text{Co}_3\text{O}_4/\text{NOCS}$. (e) Fitted O 1s peak of $\text{Co}_3\text{O}_4/\text{NOCS}$ and NOCS.

spheres are prepared via polymerization, pyrolysis and removal of SiO_2 templates. The use of ultra-small SiO_2 as a hard template can create abundant nanopores not only on the surfaces but also in the interior of the carbon materials [40]. This, in turn, helps create a high density of catalytic active sites and Co_3O_4 pinning sites in the next procedure. Then, the prepared carbon spheres were loaded with Co(II) ions (Co(II)-NOCS) and annealed at 300°C with a heating rate of 3°C min^{-1} under N_2 for 2 h to generate $\text{Co}_3\text{O}_4/\text{NOCS}$. The XRD patterns of the $\text{Co}_3\text{O}_4/\text{NOCS}$ and NOCS are provided in Fig. 1b. The two amorphous carbon peaks in $\text{Co}_3\text{O}_4/\text{NOCS}$ are observed as those in NOCS ca. 24° and 44° , respectively, which are associated with the (002) plane of graphitized carbon and (100) reflection of graphitic sheets. The position of the (002) plane in the $\text{Co}_3\text{O}_4/\text{NOCS}$ slightly shifted to the right, indicating that the decoration of Co_3O_4 nanoparticles modified the structure of the carbon support. The other marked peaks in the pattern of the $\text{Co}_3\text{O}_4/\text{NOCS}$ correspond well with those of the standard patterns of Co_3O_4 (JCPDS no. 43–1003), confirming the presence of Co_3O_4 nanomaterials in NOCS. The amount of Co_3O_4 in $\text{Co}_3\text{O}_4/\text{NOCS}$ was estimated by comparing the weight of the residue that remaining in $\text{Co}_3\text{O}_4/\text{NOCS}$ at the end of TG analysis in air with that of NOCS (Fig. 1c). The residue of the NOCS showed a value of nearly zero, while the Co_3O_4 NPs/NOCS gave ca. 10.2 wt% residue, which can be ascribed to Co_3O_4 , as NOCS should be lost in the form of CO_2 during calcination. According to the above analysis, the amount of Co_3O_4 NPs in $\text{Co}_3\text{O}_4/\text{NOCS}$ is determined to be 10.2 wt%. In addition,

it is worth noting from the TG curves that the presence of Co_3O_4 in NOCS facilitated the thermal decomposition of the materials. While NOCS is stable up to ca. 480°C in air and decomposes in the range of $480\text{--}670^\circ\text{C}$, the presence of Co_3O_4 lowers the decomposition temperature to ca. 300°C . The decreased thermal stability indirectly indicated the successful decoration of the surface of NOCS with Co_3O_4 . The oxygen and nitrogen contents of $\text{Co}_3\text{O}_4/\text{NOCS}$ determined by elemental analyzers are 14.24 wt% and 5.77 wt% (Table S1), respectively, which are 11.37 wt% and 8.60 wt% in NOCS, respectively. The increase in oxygen and decrease in nitrogen content confirm the successful decoration of metal oxides on the carbon structure.

XPS analysis was conducted to determine the oxidation state and surface element components of $\text{Co}_3\text{O}_4/\text{NOCS}$. The survey scan spectra of the NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ reveal the presence of O, N and C in both samples and Co in the $\text{Co}_3\text{O}_4/\text{NOCS}$ (Fig. S1). The O peak increases significantly when Co is present, demonstrating the presence of an oxide material in the hybrid. The carbon species on the surface of $\text{Co}_3\text{O}_4/\text{NOCS}$ and NOCS can be divided into three peaks (Fig. S2) that can be attributed to $\text{C}=\text{C}$, $\text{C}-\text{O}/\text{C}-\text{N}$ and $\text{C}=\text{O}$ [45]. All the fitted C 1s peaks in the two samples are located at consistent binding energies, demonstrating the small influence of entangled Co_3O_4 on the carbon texture. As shown in Fig. 1d, The main peaks of Co 2p and the corresponding shake-up satellites are located at 780 eV and 795 eV, representing Co $2p_{3/2}$ and Co $2p_{1/2}$, respectively. The Co $2p_{3/2}$ peak was further divided into

779.1 eV and 780.9 eV, which correspond to Co^{2+} and Co^{3+} . The two spin-orbit doublets of Co $2p_{1/2}$ were centered at 794.4 eV and 796.5 eV. The above results also confirm the presence of Co_3O_4 , which matches well with the reported paper [34]. Subsequently, the O content (Fig. 1e) of $\text{Co}_3\text{O}_4/\text{NOCS}$ can be fitted to Co–O (529.9 eV) [46], carbonyl groups (O-1, 531.3 eV), C–O ether groups (O-2, 532.7 eV) and carboxyl groups (O-3, 533.9 eV) [47], demonstrating the presence of O in both the carbon matrix and metal oxide.

To investigate the effect of the supported Co_3O_4 on the specific surface area and pore structure of this compound, nitrogen porosimetry was carried out (Fig. 2a). The N_2 adsorption–desorption curves of both the NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ give a type IV isotherm, which is typically characteristic of mesoporous materials. The surface areas of NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ determined via BET method are 817 and 450 $\text{m}^2 \text{g}^{-1}$, respectively. The significant decrease in the BET surface area after Co_3O_4 decoration is attributed to Co_3O_4 formation inside the pores of the NOCS, confirming that Co_3O_4 NPs are successfully formed in the porous structure of NOCS. Although the major pore sizes for NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ are both approximately 12 nm generated by SiO_2 templates, the pore volume (with a pore width of 12 nm) of the $\text{Co}_3\text{O}_4/\text{NOCS}$ is obviously smaller than that of the NOCS, further confirming that a large number of pores is occupied by Co_3O_4 NPs (Fig. 2b). To further analyze the influence of the supported Co_3O_4 on the pore structure, the incremental pore volumes in the two materials were compared (Fig. 2c). The volumes of the pores with a diameter of 12 nm decreased from 0.08 $\text{cm}^3 \text{g}^{-1}$ in the NOCS to 0.05 $\text{cm}^3 \text{g}^{-1}$ in the $\text{Co}_3\text{O}_4/\text{NOCS}$, demonstrating the presence of Co_3O_4 NPs in the pores of NOCS.

The morphologies and structure of NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ were examined by FE-SEM, TEM and HRTEM. The SEM image of the NOCS possesses an obvious spherical shape with abundant pores on the surface (Fig. 3a). TEM images (Fig. 3b, 3c) demonstrate that the pores are also evenly distributed inside the material. The TEM image at higher magnification (Fig. S3) confirms the evenly distributed mesopores in NOCS. In Fig. 3d, only a few brightened spots associated with Co_3O_4 species are observed on the surface of $\text{Co}_3\text{O}_4/\text{NOCS}$, indirectly demonstrating that most Co_3O_4 NPs suc-

cessfully grow in the pores of NOCS. TEM images further demonstrate that Co_3O_4 nanoparticles are well dispersed in the NOCS (Fig. 3e, 3f). HRTEM of $\text{Co}_3\text{O}_4/\text{NOCS}$ reveals that the sizes of the particles are less than 5 nm (Fig. 3g). The lattice fringe spacing of 0.467 nm corresponds to the (111) planes of Co_3O_4 , which again demonstrates the successful formation of Co_3O_4 over NOCS (Fig. 3h). Fig. 3i shows the selected-area electron diffraction (SAED) pattern of $\text{Co}_3\text{O}_4/\text{NOCS}$, which exhibits well-defined diffraction rings. The bright rings indicate the polycrystalline nature of the crystal and correspond to the (311), (400) and (440) planes of Co_3O_4 (JCPDS no. 43–1003); these results are consistent with those of XRD (Fig. 1b).

3.2. Oxygen reduction activity of $\text{Co}_3\text{O}_4/\text{NOCS}$

To evaluate the ORR performance of catalysts, CV and LSV measurements were carried out by employing a RDE or RRDE. As shown in Fig. 4a, both NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ exhibit oxygen reduction peaks in O_2 -saturated 0.1 M KOH solutions but not in N_2 -saturated electrolyte, indicating their good electrochemical activity toward the ORR. The ORR peak of $\text{Co}_3\text{O}_4/\text{NOCS}$ is located at 0.84 V vs. RHE, whereas NOCS displays a cathodic ORR peak at approximately 0.8 V vs. RHE, suggesting that the decorated Co_3O_4 plays a key role in boosting the ORR performance. For comparison, CV measurements were also conducted over $\text{Co}_3\text{O}_4/\text{CNTs}$ and commercial Pt/C (characterizations can be found in Fig. S4 and Fig. S5). As shown in Fig. S6a, the ORR peak of $\text{Co}_3\text{O}_4/\text{CNTs}$ is much more negative, with a value of 0.68 V vs. RHE, demonstrating the importance of the carbon texture. The Pt/C displays a higher ORR performance (Fig. S6b) close to that of the NOCS but still inferior to that of the $\text{Co}_3\text{O}_4/\text{NOCS}$. Subsequently, the ORR activity of $\text{Co}_3\text{O}_4/\text{NOCS}$ was further examined by LSV measurements (Fig. 4b). The half-wave potential ($E_{1/2}$) of the $\text{Co}_3\text{O}_4/\text{NOCS}$ is located at 0.82 V vs. RHE, which is more positive than that of the NOCS, $\text{Co}_3\text{O}_4/\text{CNTs}$ and even commercial Pt/C, suggesting its superior ORR activity and even better than that of Pt/C. It should be noted that when Co_3O_4 was supported on CNTs, the resulting catalyst exhibited much lower ORR activity than $\text{Co}_3\text{O}_4/\text{NOCS}$. Moreover, the activity of $\text{Co}_3\text{O}_4/\text{NOCS}$ is also superior to that of NOCS, suggesting that

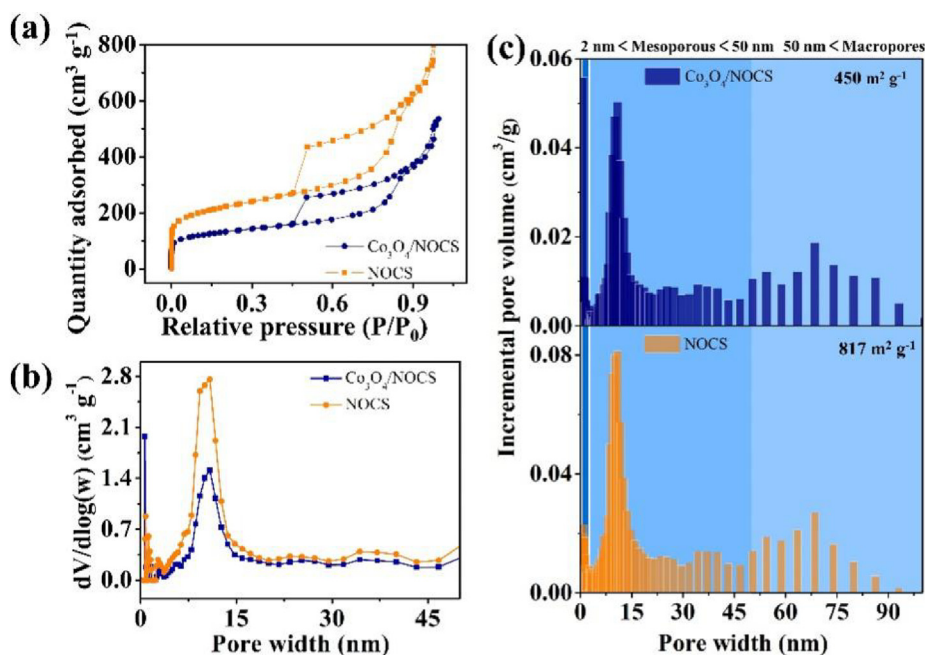


Fig. 2. (a) N_2 adsorption–desorption isotherms of NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$. (b) and (c) Pore volume distribution graphs.

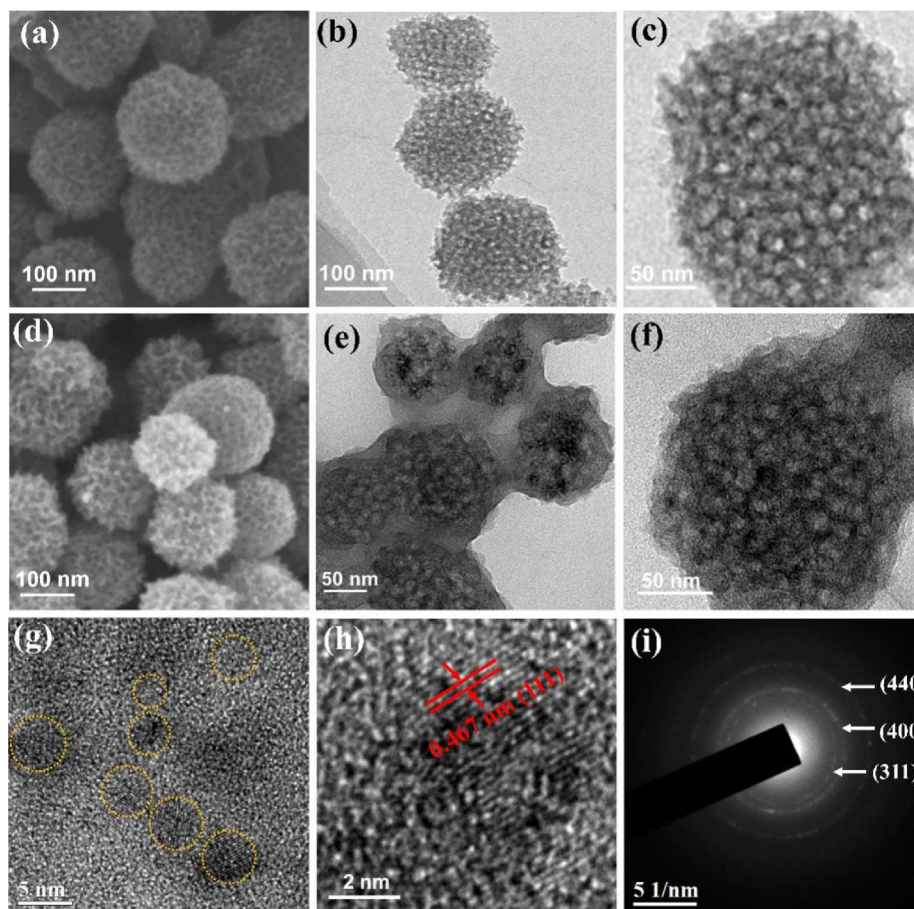


Fig. 3. SEM (a) and TEM (b–c) images of the N,O-codoped carbon spheres. SEM (d) and TEM (e–f) images of $\text{Co}_3\text{O}_4/\text{NOCS}$. (g–h) HRTEM images of $\text{Co}_3\text{O}_4/\text{NOCS}$. (i) Selected area electron diffraction (SAED) pattern of $\text{Co}_3\text{O}_4/\text{NOCS}$.

Co_3O_4 nanoparticles are necessary for boosting the ORR activity of this hybrid. The above results suggest that a synergistic effect toward the ORR is present between the Co_3O_4 NPs and NOCS.

Fig. 4c shows the LSV curves of $\text{Co}_3\text{O}_4/\text{NOCS}$ at different rotation speeds. Based on the Koutecky-Levich plots (Fig. 4d) obtained from RDE polarization curves (Fig. 4c), the electron transfer number (n) of $\text{Co}_3\text{O}_4/\text{NOCS}$ is calculated to be close to 4, indicating that this hybrid obeys first-order reaction kinetics. The value of n for the NOCS was also calculated (Fig. S7), which was found to be $3.94 \sim 4$, demonstrating that both samples possess a four-electron pathway. The Tafel plots of the $\text{Co}_3\text{O}_4/\text{NOCS}$ exhibit a Tafel slope of 78 mV dec^{-1} , which is smaller than that of NOCS (Fig. S8), suggesting the better kinetic activity of the former toward the ORR. Next, to evaluate the oxygen reduction pathways and hydrogen peroxide yield of this electrocatalyst, RRDE measurements were employed (Fig. S9). As shown in Fig. 4e, the $\text{Co}_3\text{O}_4/\text{NOCS}$ catalyst exhibits an electron transfer number of $3.88 \sim 3.94$ in the potential range of $0.2 \text{ V} \sim 0.8 \text{ V}$ vs. RHE, which is comparable to that of commercial Pt/C, which has a value of $3.79 \sim 4.0$ in the same potential range. Moreover, based on the calculation, the H_2O_2 production yield for $\text{Co}_3\text{O}_4/\text{NOCS}$ is always lower than 3.92% in the potential range of $0.2 \sim 0.8 \text{ V}$ vs. RHE. The above results further demonstrate that $\text{Co}_3\text{O}_4/\text{NOCS}$ undergoes four-electron transfer during the ORR process, which is identical to the calculated value based on the K-L equation in Fig. 4d. To obtain further insight into the effect of the confined Co_3O_4 on the electrochemical activity of the hybrid, the compound was immersed in a $0.5 \text{ M H}_2\text{SO}_4$ acid solution for 12 h to remove the Co_3O_4 nanoparticles. After acid etching, hybrid displays a relatively lower electrocatalytic activity than that of the

$\text{Co}_3\text{O}_4/\text{NOCS}$, whose $E_{1/2}$ obviously shifts in the negative direction, as shown in Fig. 4f. This analysis confirms again that the synergy between the Co_3O_4 nanoparticles and NOCS improves the activity of the compound.

Stability is crucial for the material's practical application. As displayed in Fig. 5a, the $\text{Co}_3\text{O}_4/\text{NOCS}$ exhibits outstanding stability during the chronoamperometric test at 0.82 V for 7200 s, with a current density retention of 95.48%, whereas the value for the NOCS is 92.3% under the same conditions. In addition, commercial Pt/C was used for another comparison, and the current density shows a significant decrease with a retention of only 57.14%. The excellent stability of this two-phase composite could be due to the unique structure of the confined growth of Co_3O_4 NPs in the abundant pores of the NOCS; this growth can prevent the nanomaterial from agglomerating during the electrochemical reaction. Subsequently, to evaluate whether the catalysts can be potential candidates for methanol fuel cells, the methanol tolerance of this composite was evaluated. As shown in Fig. 5b, both $E_{1/2}$ and the limited-diffusion current density based on LSV measurements show no appreciable variation in 0.1 M KOH or in a mixed electrolyte of 0.1 M KOH and 1 M methanol. The chronoamperometric test also confirms this result. After the introduction of 1 M methanol into the electrolyte, the current density of the Pt/C electrode drops sharply and finally becomes positive because of methanol oxidation on the Pt/C (Fig. 5c). In contrast, no obvious change can be observed for $\text{Co}_3\text{O}_4/\text{NOCS}$. The above results demonstrate that $\text{Co}_3\text{O}_4/\text{NOCS}$ has good tolerance toward methanol crossover and can be potentially applied in methanol fuel cells.

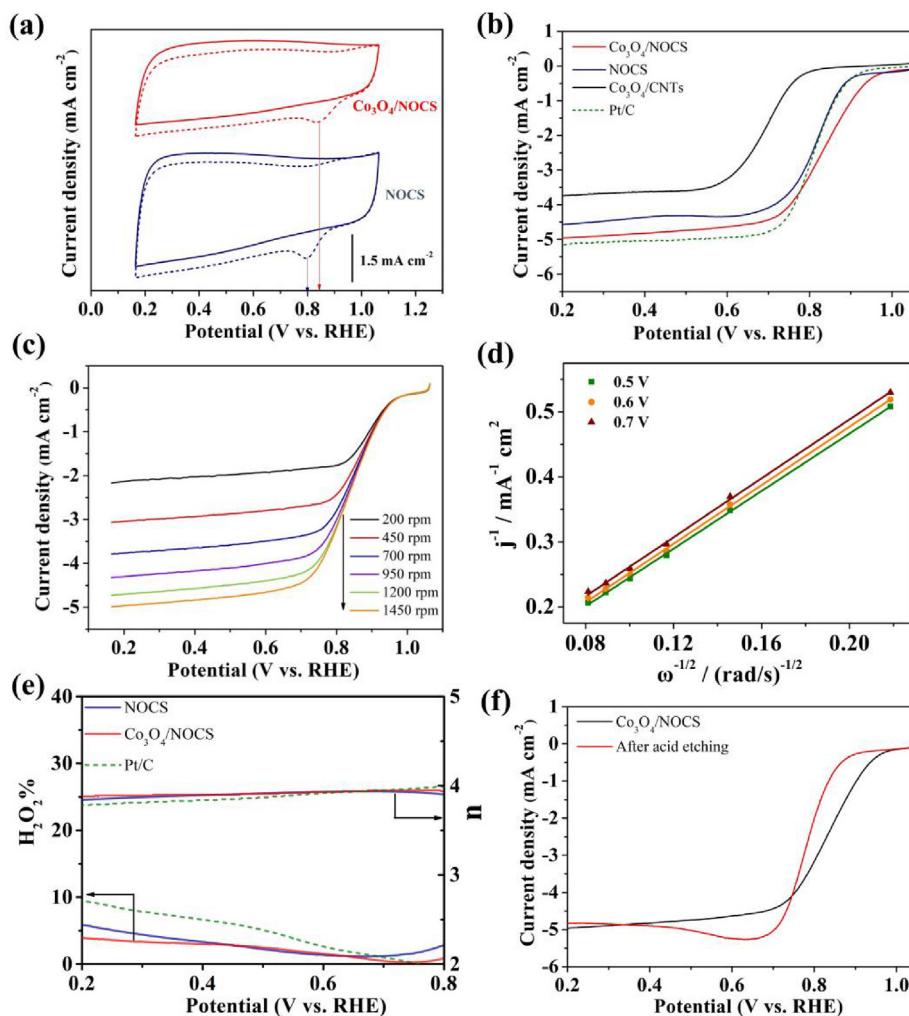


Fig. 4. (a) Cyclic voltammograms of the NOCS and $\text{Co}_3\text{O}_4/\text{NOCS}$ in 0.1 M KOH solution. Dotted line: O_2 -saturated electrolyte; solid line: N_2 -saturated electrolyte. (b) LSV curves of various catalysts in 0.1 M KOH solution with a RDE spinning at 1600 rpm. (c) LSV curves of $\text{Co}_3\text{O}_4/\text{NOCS}$ using an RDE at different rotation rates. (d) Koutecky-Levich plots for $\text{Co}_3\text{O}_4/\text{NOCS}$ sample. (e) The amount of H_2O_2 generated and number of electrons over $\text{Co}_3\text{O}_4/\text{NOCS}$, NOCS and Pt/C. (f) LSV curves of $\text{Co}_3\text{O}_4/\text{NOCS}$ before and after acid etching with a rotation rate of 1600 rpm.

3.3. The ORR activity of $\text{Co}_3\text{O}_4/\text{NOCS}$ in a flexible solid-state Zn-air battery

To demonstrate its excellent ORR ability in a battery and its practical application potential, $\text{Co}_3\text{O}_4/\text{NOCS}$ was employed as the catalyst for an air electrode (cathode) in a primary solid zinc-air battery (ZAB). The structure of our ZAB is displayed in Fig. 6a, which consists of a Zn foil, a solid-state electrolyte, and an air electrode. The characterizations of the solid-state electrolyte can be found in Fig. S10 and S11. The morphologies of the gel can be found in Fig. S10. The swelling capacity of the gel increased significantly over time. During the 66-hour test, the weight of the gel after the adsorption electrolyte reached 3.85 times that before adsorption (Fig. S11a). The ionic conductivity of the gel was calculated to be 0.39 S cm^{-1} (Fig. S11b).

As seen in Fig. 6b, the open-circuit voltage (OCV) is approximately 1.402 V, which is higher than that of a commercial Pt/C-catalyzed ZAB (1.386 V) and NOCS-catalyzed ZAB (1.219 V). The instantaneous OCV was also measured by a multimeter (Fig. S12). Fig. 6c displays the performance of ZAB driven by $\text{Co}_3\text{O}_4/\text{NOCS}$. Pt/C and NOCS were chosen as a comparison. The power density is 227 mW cm^{-2} , far higher than that of NOCS-catalyzed ZAB (195 mW cm^{-2}) and Pt/C-catalyzed ZAB (133 mW

cm^{-2}), indicating its excellent ORR performance. This result is also superior to those of previously reported catalysts (Table S2). The rate performance of the ZAB at different current densities was investigated by a galvanostatic discharging test (Fig. 6d). The $\text{Co}_3\text{O}_4/\text{NOCS}$ -catalyzed ZAB exhibits good rate performance without appreciable voltage loss while recovering the discharge at 10 mA cm^{-2} , suggesting its good ORR kinetics. On the contrary, the ZAB based on the NOCS-catalyzed air electrode shows unsatisfactory rate performance, fluctuating noticeably while discharging at 10 mA cm^{-2} and 25 mA cm^{-2} , with an obvious voltage loss while recovering the discharge at 10 mA cm^{-2} . This result also confirms that the decoration of Co_3O_4 into NOCS can improve the stability of the NOCS catalyst. Subsequently, the stability of the battery was also confirmed by galvanostatic discharging at 5 mA cm^{-2} . As displayed in Fig. 6e, no obvious fluctuation was observed during the test range of 80 h, again indicating the excellent stability of this composite toward the ORR. The $\text{Co}_3\text{O}_4/\text{NOCS}$ -based air electrode before and after discharging was analyzed by XPS, as shown in Fig. S13. The chemical state of Co 2p shows no appreciable change, whereas the binding energy of N 1s and O 1s exhibit some shift. This result can explain the better stability of the $\text{Co}_3\text{O}_4/\text{NOCS}$ compared with the NOCS catalyst.

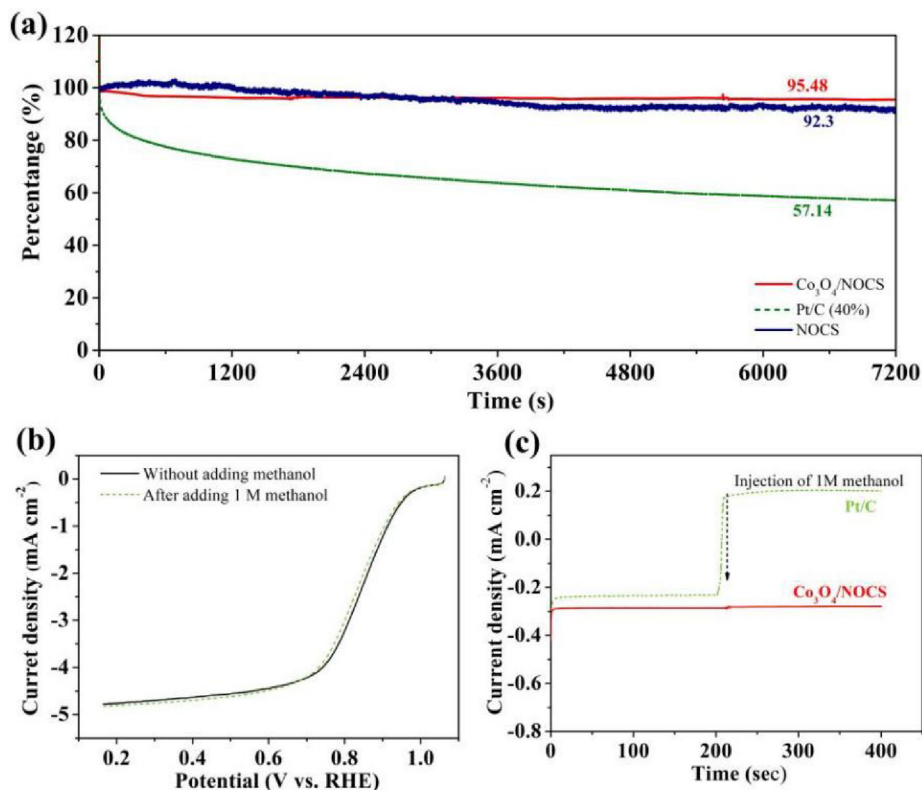


Fig. 5. (a) Stability test of various catalysts using a RDE with a spinning rate of 1200 rpm. (b) LSV curves of Co₃O₄/NOCS before and after the injection of methanol over RDE with a rotation rate of 1200 rpm. (c) Changes in current density for different catalysts with 1 M methanol injected.

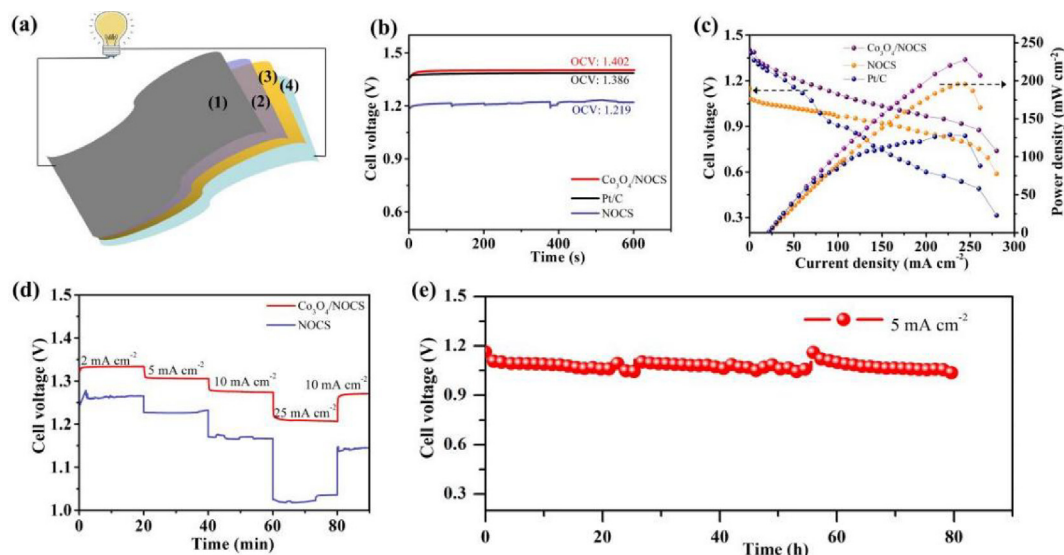


Fig. 6. (a) The structure of the flexible Zn-air battery: 1. gas diffusion layer; 2. cathode; 3. solid-state electrolyte; 4. Zn-foil. (b) Open-circuit voltage of Co₃O₄/NOCS, NOCS and Pt/C. (c) Polarization and power density curves of the primary zinc-air batteries based on Co₃O₄/NOCS, NOCS and Pt/C as the air electrodes. (d) Galvanostatic discharge curves with Co₃O₄/NOCS and NOCS catalysts at different current densities. (e) Stability test of the zinc-air battery employing Co₃O₄/NOCS as the ORR catalyst.

3.4. ORR catalytic mechanism analysis for Co₃O₄/NOCS

The above results demonstrate that Co₃O₄/NOCS has superior ORR performance and satisfactory stability. The enhanced catalytic activity is attributed from the well match of Co₃O₄ NPs and the N, O-codoped carbon. As schematically illustrated in Fig. 7a, the large specific surface area of the N,O-doped carbon spheres provides abundant active sites toward the ORR. Moreover, the large num-

bers of pores in the NOCS also endow Co₃O₄ with dispersed growth and exposed active sites. Therefore, large numbers of exposed Co³⁺ cations regarded as active sites can act as donor-acceptor reduction sites, with acceptor character by capturing electrons from the electrode and donor electrons to the electrolyte [48]. In addition, the electrons transported from the anode can be shuttled freely along the conducting network of carbon spheres and subsequently transfer to the surface of Co₃O₄ NPs to reduce O₂.

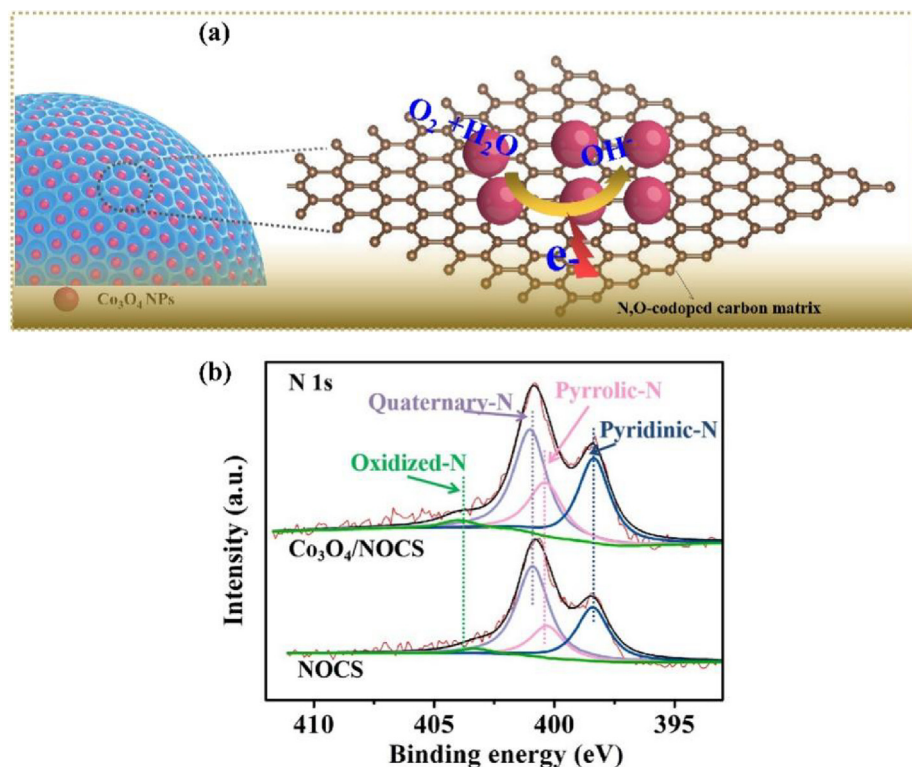


Fig. 7. (a) The possible reaction mechanism of Co₃O₄/NOCS toward the ORR. (b) Deconvoluted N 1s XPS peaks of NOCS and Co₃O₄/NOCS.

Additionally, as shown in Fig. 7b, the binding energy of quaternary N and oxidized N in Co₃O₄/NOCS shows a positive shift compared with those in NOCS, suggesting electronic interactions between the carbon support and metal oxides, which can greatly influence the adsorption energy of intermediate products during the ORR and thus optimize the ORR activity [37,49]. Regarding the improved excellent stability, the Co₃O₄ formation in the abundant pores of the NOCS can prevent them from agglomerating during the electrochemical reaction, thus rendering materials with excellent long-term stability.

4. Conclusion

In summary, a highly efficient electrochemical catalyst was developed, where Co₃O₄ nanoparticles were entrapped in the uniform mesopores of N,O-codoped carbon spheres. The hybrid material Co₃O₄/NOCS displays improved ORR performance and stability, superior to that of NOCS and commercial Pt/C. The fabricated primary solid Zn-air battery receives a high power density of 227 mW cm⁻², better than those of many Co₃O₄/C catalyzed zinc-air batteries including Co₃O₄ anchored on N-doped reduced graphene oxide (47 mW cm⁻²) [50], Co₃O₄/N-doped carbon nanofibers (75 mW cm⁻²) [34] and Co₃O₄/N-doped CNTs (112 mW cm⁻²) [32] in the literature. This result again confirms the excellent ORR activity of Co₃O₄/NOCS. The enhanced electrocatalytic activity of Co₃O₄/NOCS compared to each of the single parts could be ascribed to the special structure of the composites, where the carbon matrix with abundant mesopores and metal oxides entangled into the pores both expose abundant active sites. Moreover, the electronic interaction between the carbon support and metal oxides, confirmed by XPS, also boosted the ORR activity. In addition, the stability of the compound was also significantly improved due to this structure. In view of the excellent performance of this catalyst, the resulting material has the potential for applications in energy storage and conversion devices. The synthetic method can be

extended for the development of other electrocatalysts for the ORR and other reactions.

CRediT authorship contribution statement

Yan Liu: Conceptualization, Methodology, Data curation, Writing - original draft. **Tao Zhang:** Methodology. **Yu E. Duan:** Methodology. **Xin Dai:** Investigation. **Qiang Tan:** Investigation. **Yuanzhen Chen:** Investigation. **Yongning Liu:** Conceptualization, Methodology, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the Analysis and Test Center of Xi'an Jiaotong University for its help with XPS analyses. This work is financially supported by the National Natural Science Foundation of China (No. 51602246, 21603171).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2021.07.040>.

References

- [1] Q. Tan, T. Qu, C.-Y. Shu, Y. Liu, Y. He, W. Zhai, S.-w. Guo, L. Liu, Y.-N. Liu, High-performance polymer fiber membrane based direct methanol fuel cell system with non-platinum catalysts, *ACS Sustain. Chem. Eng.* 7 (20) (2019) 17145–17153.

- [2] Y. Wang, D.F. Ruiz Diaz, K.S. Chen, Z. Wang, X.C. Adroher, Materials, technological status, and fundamentals of PEM fuel cells - a review, *Mater. Today* 32 (2020) 178–203.
- [3] H.-F. Wang, C. Tang, Q. Zhang, A review of precious-metal-free bifunctional oxygen electrocatalysts: Rational design and applications in Zn-air batteries, *Adv. Funct. Mater.* 28 (46) (2018) 1803329, <https://doi.org/10.1002/adfm.201803329>.
- [4] W.-Z. Cheng, J.-L. Liang, H.-B. Yin, Y.-J. Wang, W.-F. Yan, J.-N. Zhang, Bifunctional iron-phthalocyanine metal-organic framework catalyst for ORR, OER and rechargeable zinc-air battery, *Rare Metals* 39 (7) (2020) 815–823.
- [5] A. Sarapu, E. Kibena-Pöldsepp, M. Borghei, K. Tammeveski, Electrocatalysis of oxygen reduction on heteroatom-doped nanocarbons and transition metal-nitrogen-carbon catalysts for alkaline membrane fuel cells, *J. Mater. Chem. A* 6 (3) (2018) 776–804.
- [6] L. Yang, X. Zeng, W. Wang, D. Cao, Recent progress in MOF-derived, heteroatom-doped porous carbons as highly efficient electrocatalysts for oxygen reduction reaction in fuel cells, *Adv. Funct. Mater.* 28 (7) (2018) 1704537, <https://doi.org/10.1002/adfm.201704537>.
- [7] F. Liu, X. Zhang, X. Zhang, L. Wang, M. Liu, J. Zhang, Dual-template strategy for electrocatalyst of cobalt nanoparticles encapsulated in nitrogen-doped carbon nanotubes for oxygen reduction reaction, *J. Colloid Interface Sci.* 581 (2021) 523–532.
- [8] L. Qu, Y. Liu, J.-B. Baek, L. Dai, Nitrogen-doped graphene as efficient metal-free electrocatalyst for oxygen reduction in fuel cells, *ACS Nano* 4 (3) (2010) 1321–1326.
- [9] W. Zhang, H. Sun, Z. Zhu, R. Jiao, P. Mu, W. Liang, A.n. Li, N-doped hard carbon nanotubes derived from conjugated microporous polymer for electrocatalytic oxygen reduction reaction, *Renew. Energy* 146 (2020) 2270–2280.
- [10] Z. Ma, S. Dou, A. Shen, L. Tao, L. Dai, S. Wang, Sulfur-doped graphene derived from cyclized lithium-sulfur batteries as a metal-free electrocatalyst for the oxygen reduction reaction, *Angew. Chem.-Int. Edit.* 54 (6) (2015) 1888–1892.
- [11] W. Qi, J. Niu, Z. Huang, Z. Chen, X. Chen, F. Chen, J. Tuo, Mechanisms insight into oxygen reduction reaction on sulfur-doped Fe-N-2 graphene electrocatalysts, *Int. J. Hydrog. Energy* 45 (1) (2020) 521–530.
- [12] W. Liu, J. Zhang, Z. Bai, G. Jiang, M. Li, K. Feng, L. Yang, Y. Ding, T. Yu, Z. Chen, A. Yu, Controllable urchin-like NiCo₂S₄ microsphere synergized with sulfur-doped graphene as bifunctional catalyst for superior rechargeable Zn-air battery, *Adv. Funct. Mater.* 28 (11) (2018) 1706675, <https://doi.org/10.1002/adfm.201706675>.
- [13] K. Yuan, S. Sfaelou, M. Qiu, D. Lützenkirchen-Hecht, X. Zhuang, Y. Chen, C. Yuan, X. Feng, U. Scherf, Synergetic contribution of boron and Fe-Nx species in porous carbons toward efficient electrocatalysts for oxygen reduction reaction, *ACS Energy Lett.* 3 (1) (2018) 252–260.
- [14] H. Fei, R. Ye, G. Ye, Y. Gong, Z. Peng, X. Fan, E.L.G. Samuel, P.M. Ajayan, J.M. Tour, Boron- and nitrogen-doped graphene quantum dots/graphene hybrid nanoplatelets as efficient electrocatalysts for oxygen reduction, *ACS Nano* 8 (10) (2014) 10837–10843.
- [15] L. Han, X. Cui, Y. Liu, G. Han, X. Wu, C. Xu, B. Li, Nitrogen and phosphorus modification to enhance the catalytic activity of biomass-derived carbon toward the oxygen reduction reaction, *Sustain. Energ. Fuels* 4 (6) (2020) 2707–2717.
- [16] S.C. Huang, Y.Y. Meng, Y.F. Cao, S.M. He, X.H. Li, S.F. Tong, M.M. Wu, N-, O- and P-doped hollow carbons: Metal-free bifunctional electrocatalysts for hydrogen evolution and oxygen reduction reactions, *Appl. Catal. B-Environ.* 248 (2019) 239–248.
- [17] L.J. Yang, J.L. Shui, L. Du, Y.Y. Shao, J. Liu, L.M. Dai, Z. Hu, Carbon-based metal-free ORR electrocatalysts for fuel cells: Past, present, and future, *Adv. Mater.* 31 (2019) 20.
- [18] C.G. Hu, Y. Lin, J.W. Connell, H.M. Cheng, Y. Gogotsi, M.M. Titirici, L.M. Dai, Carbon-based metal-free catalysts for energy storage and environmental remediation, *Adv. Mater.* 31 (2019) 14.
- [19] C.G. Hu, L.M. Dai, Doping of carbon materials for metal-free electrocatalysis, *Adv. Mater.* 31 (2019) 17.
- [20] Y. Liu, C.Y. Shu, Y. Fang, Y.Z. Chen, Y.N. Liu, Two 3D structured Co-Ni bimetallic oxides as cathode catalysts for high-performance alkaline direct methanol fuel cells, *J. Power Sources* 361 (2017) 160–169.
- [21] G. Fu, X. Yan, Y. Chen, L. Xu, D. Sun, J.-M. Lee, Y. Tang, Boosting bifunctional oxygen electrocatalysis with 3D graphene aerogel-supported Ni/MnO particles, *Adv. Mater.* 30 (5) (2018) 1704609, <https://doi.org/10.1002/adma.201704609>.
- [22] Z. Xiao, F. Hou, Y. Li, R. Zhang, G. Shen, L. Wang, X. Zhang, Q. Wang, G. Li, Confinement of Fe₂O₃ nanoparticles in the shell of N-doped carbon hollow microsphere for efficient oxygen reduction reaction, *Chem. Eng. Sci.* 207 (2019) 235–246.
- [23] R. Zhao, J. Chen, Z. Chen, X. Jiang, G. Fu, Y. Tang, W. Jin, J.-M. Lee, S. Huang, Atomically dispersed CoN₄/B, N-C nanotubes boost oxygen reduction in rechargeable Zn-air batteries, *ACS Appl. Energy Mater.* 3 (5) (2020) 4539–4548.
- [24] I.S. Amiin, Z. Pu, X. Liu, K.A. Owusu, H.G.R. Monestel, F.O. Boakye, H. Zhang, S. Mu, Multifunctional Mo-N/C@MoS₂ electrocatalysts for HER, OER, ORR, and Zn-air batteries, *Adv. Funct. Mater.* 27 (44) (2017) 1702300, <https://doi.org/10.1002/adfm.201702300>.
- [25] L. Chen, Li-Li Cui, Z. Wang, X. He, W. Zhang, T. Asefa, Co₂FeS₈/N, S-doped carbons derived from Fe-Co/S-bridged polyphthalocyanine: Efficient dual-function air-electrode catalysts for rechargeable Zn-air batteries, *ACS Sustain. Chem. Eng.* 8 (35) (2020) 13147–13158.
- [26] P. Ganesan, M. Prabu, J. Sanetuntikul, S. Shanmugam, Cobalt sulfide nanoparticles grown on nitrogen and sulfur codoped graphene oxide: An efficient electrocatalyst for oxygen reduction and evolution reactions, *ACS Catal.* 5 (2015) 3625–3637.
- [27] L. Song, J. Tang, T. Wang, C. Wu, Y. Ide, J.P. He, Y.S.K. Yamauchi, Self-supported ZIF-derived Co₃O₄ nanoparticles-decorated porous N-doped carbon fibers as oxygen reduction catalyst, *Chem. Eur. J.* 25 (2019) 6807–6813.
- [28] S.H. Peng, T.H. Chen, C.H. Lee, H.C. Lu, S.J.J. Lue, Optimal cobalt oxide (Co₃O₄): Graphene (GR) ratio in Co₃O₄/GR as air cathode catalyst for air-breathing hybrid electrolyte lithium-air battery, *J. Power Sources* 471 (2020) 9.
- [29] C. Guan, A. Sumboja, H.J. Wu, W.N. Ren, X.M. Liu, H. Zhang, Z.L. Liu, C.W. Cheng, S.J. Pennycook, J. Wang, Hollow Co₃O₄ nanosphere embedded in carbon arrays for stable and flexible solid-state zinc-air batteries, *Adv. Mater.* 29 (2017) 9.
- [30] T. Hu, Y. Wang, L. Zhang, T. Tang, H. Xiao, W. Chen, M. Zhao, J. Jia, H. Zhu, Facile synthesis of PdO-doped Co₃O₄ nanoparticles as an efficient bifunctional oxygen electrocatalyst, *Appl. Catal. B-Environ.* 243 (2019) 175–182.
- [31] B. Tang, J. Yang, Z. Kou, L. Xu, H.L. Seng, Y. Xie, A.D. Handoko, X. Liu, Z.W. Seh, H. Kawai, H. Gong, W. Yang, Surface-engineered cobalt oxide nanowires as multifunctional electrocatalysts for efficient Zn-air batteries-driven overall water splitting, *Energy Stor. Mater.* 23 (2019) 1–7.
- [32] Z. Huang, X. Qin, G. Li, W. Yao, J. Liu, N. Wang, K. Ithiuphalap, G. Wu, M. Shao, Z. Shi, Co₃O₄ nanoparticles anchored on nitrogen-doped partially exfoliated multiwall carbon nanotubes as an enhanced oxygen electrocatalyst for the rechargeable and flexible solid-state Zn-air battery, *ACS Appl. Energ. Mater.* 2 (6) (2019) 4428–4438.
- [33] X. Liu, T.L. Jin, Z.D. Hood, C. Tian, Y. Guo, W. Zhan, Mechanochemically assisted synthesis of ruthenium clusters embedded in mesoporous carbon for an efficient hydrogen evolution reaction, *ChemElectroChem* 6 (10) (2019) 2719–2725.
- [34] L. Qiu, X. Han, Q. Lu, J. Zhao, Y. Wang, Z. Chen, C. Zhong, W. Hu, Y. Deng, Co₃O₄ nanoparticles supported on N-doped electrospinning carbon nanofibers as an efficient and bifunctional oxygen electrocatalyst for rechargeable Zn-air batteries, *Inorg. Chem. Front.* 6 (12) (2019) 3554–3561.
- [35] J. Guan, Z. Zhang, J. Ji, M. Dou, F. Wang, Hydrothermal synthesis of highly dispersed Co₃O₄ nanoparticles on biomass-derived nitrogen-doped hierarchically porous carbon networks as an efficient bifunctional electrocatalyst for oxygen reduction and evolution reactions, *ACS Appl. Mater. Interfaces* 9 (36) (2017) 30662–30669.
- [36] C. Guan, A. Sumboja, W. Zang, Y. Qian, H. Zhang, X. Liu, Z. Liu, D. Zhao, S.J. Pennycook, J. Wang, Decorating Co/CoNx nanoparticles in nitrogen-doped carbon nanorods for flexible and rechargeable zinc-air batteries, *Energy Stor. Mater.* 16 (2019) 243–250.
- [37] Y. Liu, L. Yang, B. Xie, N. Zhao, L. Yang, F. Zhan, Q. Pan, J. Han, X. Wang, J. Liu, J. Li, Y. Yang, Ultrathin Co₃O₄ nanosheet clusters anchored on nitrogen doped carbon nanotubes/3D graphene as binder-free cathodes for al-air battery, *Chem. Eng. J.* 381 (2020) 122681, <https://doi.org/10.1016/j.cej.2019.122681>.
- [38] M.S. Ahmed, B. Choi, Y.B. Kim, Development of highly active bifunctional electrocatalyst using Co₃O₄ on carbon nanotubes for oxygen reduction and oxygen evolution, *Sci Rep* 8 (2018) 10.
- [39] N. Manna, S.K. Singh, G.P. Kharabe, A. Torris, S. Kurungot, Zinc-air batteries catalyzed using Co₃O₄ nanorod-supported N-doped entangled graphene for oxygen reduction reaction, *ACS. Appl. Energ. Mater.* 4 (5) (2021) 4570–4580.
- [40] Y. Liu, T. Zhang, X. Dai, Q. Tan, Y. Chen, Y. Liu, T. Asefa, Ordered nanoporous nitrogen- and oxygen-codoped carbon nanospheres as electrocatalysts for oxygen-reduction reaction in direct methanol fuel cells, *ACS Appl. Nano Mater.* 3 (6) (2020) 5139–5148.
- [41] S. Gupta, S. Zhao, X.X. Wang, S. Hwang, S. Karakalos, S.V. Devaguptapu, S. Mukherjee, D. Su, H. Xu, G. Wu, Quaternary FeCoNiMn-based nanocarbon electrocatalysts for bifunctional oxygen reduction and evolution: Promotional role of Mn doping in stabilizing carbon, *ACS Catal.* 7 (2017) 8386–8393.
- [42] A. Pendashteh, J. Palma, M. Anderson, R. Marcilla, NiCoMnO₄ nanoparticles on N-doped graphene: Highly efficient bifunctional electrocatalyst for oxygen reduction/evolution reactions, *Appl. Catal. B-Environ.* 201 (2017) 241–252.
- [43] B. Huang, X. Hu, Y. Liu, W. Qi, Z. Xie, Biomolecule-derived N/S co-doped CNT-graphene hybrids exhibiting excellent electrochemical activities, *J. Power Sources* 413 (2019) 408–417.
- [44] Y.Y. Xu, P.L. Deng, G.D. Chen, J.X. Chen, Y. Yan, K. Qi, H.F. Liu, B.Y. Xia, 2D nitrogen-doped carbon nanotubes/graphene hybrid as bifunctional oxygen electrocatalyst for long-life rechargeable Zn-air batteries, *Adv. Funct. Mater.* 30 (2020) 8.
- [45] J. Ding, P. Wang, S. Ji, H. Wang, V. Linkov, R. Wang, N-doped mesoporous FeNx/carbon as ORR and OER bifunctional electrocatalyst for rechargeable zinc-air batteries, *Electrochimica Acta* 296 (2019) 653–661.
- [46] Q. Li, L. Wang, X. Liu, P. Yu, C. Tian, H. Fu, N-doped carbon-coated Co₃O₄ nanosheet array/carbon cloth for stable rechargeable Zn-air batteries, *Sci. China-Mater.* 62 (5) (2019) 624–632.
- [47] Z. Song, L. Li, D. Zhu, L. Miao, H. Duan, Z. Wang, W. Xiong, Y. Lv, M. Liu, L. Gan, Synergistic design of a N, O co-doped honeycomb carbon electrode and an ionogel electrolyte enabling all-solid-state supercapacitors with an ultrahigh energy density, *J. Mater. Chem. A* 7 (2) (2019) 816–826.
- [48] J. Xu, P. Gao, T.S. Zhao, Non-precious Co₃O₄ nano-rod electrocatalyst for oxygen reduction reaction in anion-exchange membrane fuel cells, *Energy Environ. Sci.* 5 (1) (2012) 5333–5339.
- [49] W.-J. Jiang, L. Gu, L. Li, Y. Zhang, X. Zhang, L.-J. Zhang, J.-Q. Wang, J.-S. Hu, Z. Wei, L.-J. Wan, Understanding the high activity of Fe–N–C electrocatalysts in

- oxygen reduction: Fe/Fe₃C nanoparticles boost the activity of Fe–Nx, *J. Am. Chem. Soc.* 138 (10) (2016) 3570–3578.
- [50] J.S. Sanchez, R.R. Maça, A. Pendashteh, V. Etacheri, V.A. de la Peña O'Shea, M. Castillo-Rodríguez, J. Palma, R. Marcilla, Hierarchical Co₃O₄ nanorods anchored on nitrogen doped reduced graphene oxide: A highly efficient bifunctional electrocatalyst for rechargeable Zn-air batteries, *Catalysis, Science & Technology* 10 (5) (2020) 1444–1457.